

Siva Kumar Reddy Surasani

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EXPERIENCE

Machine Learning Engineer | Micro.AI – Plasma Computing Group - Irving, TX, USA

July 2025 - Present

- Developed a DTW-based root cause analysis algorithm for industrial time-series sensor data, enabling anomaly detection and fault localization across multi-channel sensor streams on edge devices.
- Built industrial sensor data ingestion pipelines handling MQTT and OPC-UA message formats, parsing and normalizing real-time telemetry from heterogeneous edge sensors for downstream ML inference.
- Engineered state-machine-based control plane workflows for edge device onboarding and sensor integration, managing device registration, configuration validation, and monitoring activation on Embedded Linux.
- Deployed Edge AI inference pipelines on resource-constrained devices, optimizing model execution for low-latency, low-memory environments and integrating outputs with real-time sensor control loops.

Research Assistant | University of North Texas - Denton, TX, USA, Link: [Project](#)

Sep 2023 - May 2025

- Designed and developed an AI-powered accessibility chatbot using Python, LangChain, and OpenAI, building document-grounded conversational pipelines for WCAG-compliant guideline retrieval and Q&A.
- Implemented community detection and graph analytics algorithms on network datasets; built data pipelines for graph preprocessing, adjacency representation, and modularity-based clustering evaluation.
- Developed NLP pipelines and ML models (MLP, classification) to extract and predict material properties from scientific papers; improved prediction accuracy through feature engineering and comparative model evaluation.
- Built low-level software systems leveraging CUDA for direct memory management and high-performance computation on ARM-based GPU architectures; developed dynamic sparse graph data structures with optimized memory layouts on Embedded Linux environments.

Senior Software Engineer | Justdial - Bangalore, India, Link: [Justdial](#)

July 2022 - Aug 2023

- Built an OpenCV-based image processing tool in Python to crop QR codes from images, improving efficiency for financial service use cases.
- Maintained and scaled microservices using Python and RESTful APIs for real-time financial transactions and user-facing services.
- Built Jenkins CI/CD pipelines for containerized deployments on AWS, supporting test coverage and automated releases.
- Collaborated in an agile team, leading sprint planning and participating in daily stand-ups, retros, and code reviews.
- Documented authentication/authorization logic for REST APIs and enforced security best practices in payment workflows.
- Developed frontend dashboards in Angular and React.js to visualize summaries and support business logic validation for 10K+ users.
- Wrote technical and user documentation for deployment pipelines and API services, enabling seamless handoffs across teams.

Software Engineer | Droom - Gurugram, India, Link: [Droom](#)

Aug 2020 - June 2022

- Designed and deployed a Python-based recommendation engine using Elasticsearch and embeddings, increasing engagement by 30%.
- Developed listing-matching APIs in Python using scikit-learn for relevance scoring, improving lead quality and user traffic by 25%.
- Used MongoDB and Elasticsearch to implement product catalog search and user-specific recommendations.
- Experimented with hybrid ML workflows using TensorFlow Lite (PoC) for on-device lead scoring in vehicle listings.
- Built scalable React.js interfaces for e-commerce workflows and integrated them with REST APIs using secure session management.
- Dockerized Python-based services and integrated with CI pipelines for consistent production and staging environments.

EDUCATION

University of North Texas (Aug 2023 - May 2025)

Denton, TX, USA

Masters in Computer Science: **GPA - 4.00**

Indian Institute of Information Technology Kota (July 2016 – May 2020)

Jaipur, India

Bachelors in Electronics and Communications: **GPA - 3.40**

SKILLS

Languages & Frameworks: Python, GoLang, Flask, React.js, SQL, MongoDB, MySQL

Others: FAISS, Pandas, LangChain, TensorFlow, CUDA, Embedded Linux, RTOS, ARM Cortex, TCP/IP, UART, Prompt Engineering

PROJECTS

Grant Writing Assistant

[GrantAIdc](#)

- Designed and deployed a Flask-based Grant Writing Assistant integrating LangChain, FAISS, and Google GenAI.
- Built RESTful endpoints for tasks like summarization, query, and structured data extraction to enable PDF-based reasoning pipelines.
- Improved drafting speed by 60% through automation and intelligent content assembly using LLMs.
- Technologies: Python, Scikit-Learn, LLM, React.js, Flask, GCP, Firestore, and AWS.